Extensions of residues containing subfield groups and tori

Human Compression project

Abstract

We determine all rank-four residually weakly primitive extensions of the specified residues R10, R11 and R13 in the fixed higher-rank Suzuki paper. A torus subgroup together with its full Weyl complement determines its containing subfield groups uniquely. A second argument extracts a Boolean divisor lattice from residual weak primitivity. Neither argument assumes the source's classification of all residues.

1 Premises and residual conditions

Let $S_f = \mathrm{Sz}(2^f)$, with f odd, denote a standard subfield group of S_N, where $f \mid N$. We use the classical subgroup and conjugacy theorems for Suzuki groups, stated in the accepted preliminary papers. In particular, every proper nonsoluble subgroup is a simple subfield group. Simple subfield groups are self-normalizing, and their copies inside any given Suzuki group form one conjugacy class. The other maximal subgroups are point stabilizers and normalizers of the three cyclic torus types. A nonidentity odd-order element determines its torus uniquely. The three torus orders

$$2^f - 1, \qquad 2^f + 2^{(f+1)/2} + 1, \qquad 2^f - 2^{(f+1)/2} + 1$$

are pairwise coprime. Their normalizers have respective complements of orders $2, 4, 4$. In each case the complement contains an involution inverting the torus. Copies of the standard subfield groups intersect according to $S_a \cap S_b = S_{\gcd(a,b)}$.

The accepted torus construction supplies two compatible nonsplit families T_f^+, T_f^-, together with a common complement of order four. Their signs need not agree with the signs of the square-root formulas at every exponent. What matters here is

$$T_a^\epsilon \cap T_b^\epsilon = T_{\gcd(a,b)}^\epsilon, \qquad \gcd(|T_a^\epsilon|, |T_b^\epsilon|) = |T_{\gcd(a,b)}^\epsilon|. \tag{1}$$

For $d \mid f$, the two bottom orders divide the two different upper orders. These are reused structural premises, proved in the accepted paper *Subfield and torus geometries*.

Write $G_J = \bigcap_{j \in J} G_j$ and $G_\varnothing = G$. Residual weak primitivity means that for each proper $J \subset I$ some $G_{J \cup \{i\}}$ is maximal proper in G_J. It implies

$$G_{J \cup \{i\}} < G_J \quad (i \notin J). \tag{2}$$

Indeed, equality would imply $G_{I \setminus \{i\}} \leq G_i$; the last panel would then equal G_I, contrary to maximal proper inclusion. The arguments below only need this group condition.

2 Uniqueness of subfield overgroups

Lemma 1. *A fixed simple subfield group is contained in at most one subfield group of any prescribed degree. The same assertion holds for $H = C_m : C_c$ when $m > 1$ is odd and C_c is the full torus complement: $c = 2$ in a split torus, or $c = 4$ in a nonsplit torus.*

Proof. The first assertion follows from conjugacy of subfield copies and self-normalization. More generally, suppose $H \leq S_f$ and a conjugate gS_fg^{-1} also contains H. If the copies of H inside S_f are conjugate, adjust g by an element of S_f to normalize H. When $N_G(H) \leq S_f$, this forces $gS_fg^{-1} = S_f$.

For the stated torus subgroup, C_m is characteristic: it consists of the identity and all odd-order elements of H. Therefore its normalizer is contained in the normalizer $T : C_c$ of its unique torus. Remove the complement component of a normalizing element, leaving $h \in T$. If $z \in C_c$ is the involution, then $hzh^{-1} = h^2z$ under a choice of conjugation convention. Normalization of H implies $h^2 \in C_m$. Since $|T|$ is odd, this implies $h \in C_m$. Hence $N_G(H) = H$.

Copies of H in a Suzuki group are conjugate: the odd subgroup selects a unique torus type, the cyclic torus has a unique subgroup of order m, and its complements are conjugate in the odd-order torus extension. This proves the second assertion by the preceding transporter argument. The use of the full complement is essential; the assertion does not cover $C_m : C_2$ inside a nonsplit torus. □

Lemma 2. *Suppose two simple subfield groups of degrees a, b contain the same $H = C_m : C_c$ of Lemma 1. In the nonsplit case assume $m > 5$. Their intersection is a simple subfield group of degree $d = \gcd(a, b)$.*

Proof. The split type persists in every odd-degree subfield containing it: a nonsplit occurrence would make its odd order divide both $2^N - 1$ and $2^{2N} + 1$, whose gcd is one. In the split case $m \mid \gcd(2^a - 1, 2^b - 1) = 2^d - 1$. Thus H has a standard copy in S_d. Here $d > 1$, since $m > 1$. In the nonsplit case H is nonsplit in both groups, since its complement has order four. The identity

$$\gcd(2^{2a} + 1, 2^{2b} + 1) = 2^{2d} + 1$$

holds because a/d and b/d are odd. Thus m divides the product of the two nonsplit orders at d. It divides just one of them: each bottom order divides a different upper order at a, whereas m divides only the upper order containing its torus. Coprimality of the upper orders excludes any factor from the other bottom order. The compatible torus family therefore places a copy of H in S_d. Its action agrees with that of H: the order-four action is raising to $\pm 2^d$, and an odd multiple of d gives this action or its inverse modulo m. Reversing a complement generator identifies the two presentations. Since the only nontrivial nonsplit order at $d = 1$ is 5, the assumption $m > 5$ makes $d > 1$.

Conjugate the copy of H in $S_d \leq S_a$ onto the given one inside the first group. In these coordinates the standard S_b is an overgroup of H. Lemma 1 identifies it with the second group. Their intersection is now the standard S_d. □

Lemma 3. *Let $G_0 = S_{N/c}$ be maximal in S_N, where c is prime. Suppose a proper simple subfield group K shares an embedded simple subfield with G_0, and $G_0 \cap K = S_{N/(ca)}$, where a is prime and the common group is simple. If $K \not\leq G_0$, then $a \neq c$ and $K = S_{N/a}$ after simultaneous standardization.*

Proof. The common simple group and Lemma 1 make both groups standard. If v is the degree of K, then $v \mid N$, $v \nmid N/c$, and $\gcd(v, N/c) = N/(ca)$. Since N/c lacks just one occurrence of the prime c, the first two conditions force $v_c(v) = v_c(N)$. Consequently $a \neq c$, as otherwise the gcd would have to lack a second occurrence of c. At a the gcd equality forces $v_a(v) = v_a(N) - 1$; at every other prime different from c it forces the full valuation in N. Thus $v = N/a$. This proof allows a and c to divide the common base exponent. □

3 A divisor lattice forced by residual primitivity

Lemma 4. *Let $L = C_M : C_c$ have odd rotation order M. Let $H_i = C_{d_i} : C_c$ for $1 \le i \le r$, with a common fixed complement, and put $\gamma = \gcd(d_1, \dots, d_r)$. If these intersections satisfy residual weak primitivity, then there are distinct primes $\ell_1, \dots, \ell_r$ such that*

$$M = \gamma \prod_{i=1}^{r} \ell_i, \qquad d_i = M/\ell_i. \tag{3}$$

Primes ℓ_i may divide γ.

Proof. For $r = 1$ the conclusion is precisely maximality; assume $r \ge 2$. Every subgroup between $C_d : C_c$ and $C_M : C_c$ has the form $C_h : C_c$, $d \mid h \mid M$: it contains the complement, and removing that component from each element leaves its rotation. Thus a proper inclusion is maximal exactly when the ratio of rotation orders is prime. Successively applying residual weak primitivity gives r prime-index steps from L to the full meet. Hence $\Omega(M/\gamma) = r$, with multiplicity counted.

For each i, (2) says $\gcd(d_j : j \ne i) > \gamma$. Some prime ℓ_i therefore has

$$v_{\ell_i}(d_i) < \min_{j \ne i} v_{\ell_i}(d_j).$$

One prime cannot witness this inequality for two different indices. We have found r distinct primes dividing M/γ. Since its total number of prime factors is r, each occurs exactly once and there are no other prime factors. At the witnessing prime, d_i has the bottom valuation and every other d_j the top valuation. At all other primes every d_j has the valuation of M. These statements give (3), including when a witnessing prime divides γ. □

4 The torus residues R10 and R11

The R11 row calls n, n' two primes without expressly distinguishing them. The prescribed lattice forces them to be distinct: if $n = n'$, the subgroups G_{012}, G_{023} have the same cyclic rotation subgroup inside G_{02} and contain the same full complement G_{0123}. They coincide, so their meet has order $c\gamma n$ instead of the prescribed $c\gamma$. Thus the equal-prime case has no realization.

Lemma 5. *In an actual rank-three residually weakly primitive residue of the stated R10 or R11 form, write its ambient group as $\mathrm{Sz}(s)$, its two simple parabolics as $\mathrm{Sz}(t), \mathrm{Sz}(t')$, and their common simple intersection as $\mathrm{Sz}(u)$, where $u \ge 8$. Suppose the third parabolic is $C_{\gamma nn'} : C_c$, with its two corresponding intersections $C_{\gamma n} : C_c$ and $C_{\gamma n'} : C_c$, and common intersection $C_\gamma : C_c$. Then*

$$t = u^a, \qquad t' = u^b, \qquad s = u^{ab}$$

for distinct odd primes a, b. Thus these degree conditions follow from the residue and need not be supplied as extra hypotheses.

Proof. The common simple group standardizes the three field groups. Conjugation inside $\mathrm{Sz}(u)$ makes its full torus normalizer standard without disturbing these field groups. Their tori then belong to the same compatible family. Write h_f for its order at exponent f, and $u = 2^e$, so $h_e = \gamma$. If $f' > f \ge 3$ and $f \mid f'$, then h_f is a proper divisor of $h_{f'}$. Divisibility is (1). Strictness is immediate for the split family. For either nonsplit family it follows from $f' \ge 3f$ and

$$2^{3f} - 2^{(3f+1)/2} + 1 > 2^f + 2^{(f+1)/2} + 1 \quad (f \ge 3),$$

since the lower bound for an upper torus order exceeds the upper bound for a bottom torus order.

Let $t = 2^v$. If v/e were composite, an intermediate exponent $e < w < v$ would give a strict divisor chain $\gamma = h_e < h_w < h_v$. In particular h_v/γ could not be the prime n. Therefore $C_{\gamma n} : C_c$ would be strictly contained in the full torus normalizer of $\mathrm{Sz}(t)$, and would not be maximal. Residual weak primitivity at that simple parabolic would force $\mathrm{Sz}(u)$ to be maximal in $\mathrm{Sz}(t)$, which makes v/e prime, a contradiction. Thus $v/e = a$ is prime. The same argument gives $t' = u^b$ with b prime. If $a = b$, uniqueness of the containing field group would make the two parabolics equal, contrary to their prescribed intersection $\mathrm{Sz}(u)$. Hence $a \neq b$.

The degree of s is a multiple of eab. If it were larger, the four field exponents $e < ea < eab < \log_2(s)$ would give three strict torus-order increases. The ratio of the ambient torus order to γ would have at least three prime factors with multiplicity, so it could not equal nn'. The third parabolic would not be maximal. Residual weak primitivity in $\mathrm{Sz}(s)$ would then make one of $\mathrm{Sz}(u^a), \mathrm{Sz}(u^b)$ maximal. Its relative degree is respectively a multiple of b or a, with an additional factor, and is not prime. This is a contradiction. Thus $s = u^{ab}$, as asserted. □

Theorem 6. *Let $u = 2^e \geq 8$ and let a, b be distinct odd primes. Suppose a rank-four residually weakly primitive pregeometry in $\mathrm{Sz}(q)$ has a maximal parabolic $G_0 = \mathrm{Sz}(u^{ab})$ with the prescribed residue*

$$G_{01} = \mathrm{Sz}(u^a), \quad G_{03} = \mathrm{Sz}(u^b), \quad G_{02} = C_{\gamma nn'} : C_c,$$

$$G_{013} = \mathrm{Sz}(u), \quad G_{012} = C_{\gamma n} : C_c, \quad G_{023} = C_{\gamma n'} : C_c, \quad G_{0123} = C_\gamma : C_c, \tag{4}$$

where n, n' are distinct primes. For R10 put $c = 2$, $\gamma = u - 1$; for R11 put $c = 4$ and let γ be a nonsplit torus order at u. Then an odd prime d distinct from a, b satisfies

$$q = u^{abd}, \qquad G_1 = \mathrm{Sz}(u^{ad}), \qquad G_3 = \mathrm{Sz}(u^{bd}), \qquad G_{13} = \mathrm{Sz}(u^d).$$

There is a prime n'' distinct from n, n' such that

$$\begin{aligned} G_2 &= C_{\gamma nn'n''} : C_c, \\ G_{12} &= C_{\gamma nn''} : C_c, \qquad G_{23} = C_{\gamma n'n''} : C_c, \\ G_{123} &= C_{\gamma n''} : C_c. \end{aligned} \tag{5}$$

These formulas hold simultaneously after conjugation. In R10, $n'' \mid (u^d - 1)/(u - 1)$. In R11, $\gamma n''$ divides the compatible nonsplit torus order at u^d, and $\gamma nn'n''$ divides its ambient order. All remaining intersections are determined by (4)–(5).

Proof. The groups G_1, G_3 contain the simple groups G_{01}, G_{03}, so are simple subfield groups. All share G_{013}, allowing simultaneous standardization by Lemma 1. Maximality of G_0 gives $q = (u^{ab})^d$ for a prime d. Applying Lemma 3 to its two specified maximal subfield intersections gives $d \neq a, b$, and the asserted degrees and meet. Conjugate the common full torus normalizer G_{0123} inside S_e onto the chosen standard normalizer. This preserves all three standard simple groups and fixes their torus coordinates simultaneously.

Put $H = G_{02}$. If G_2 were a simple subfield group, Lemma 2 would make $G_0 \cap G_2$ a simple subfield group, contradicting $G_0 \cap G_2 = H$. In the nonsplit case its rotation order $\gamma nn'$ exceeds five even when $\gamma = 5$. This explains why one must use the entire H, rather than claim that its base rotation C_γ generates the base field.

Choose a minimal Suzuki overgroup of G_2, allowing the ambient group. Since G_2 is not itself a Suzuki group, it lies in a maximal subgroup of this overgroup. Minimality rules out the subfield type, so it lies in a point stabilizer or torus normalizer. A point stabilizer

cannot contain H: its quotient by its normal two-subgroup is cyclic of odd order, whereas the involution in H inverts a nontrivial odd rotation. The images of that involution and its inverse action would force this rotation to equal its inverse. It cannot disappear in the two-subgroup either. The wrong torus types are excluded by coprimality. The unique torus containing the odd part of H is therefore also the torus for G_2. Thus $G_2 = C_M : C_c$, with the same complement as H.

Its three parabolic intersections with G_0, G_1, G_3 all contain the complement from G_{0123}. Their full meet is $C_\gamma : C_c$. Lemma 4 says that M/γ is the product of three distinct primes, one omitted at each parabolic. The prescribed orders of G_{02}, G_{012}, G_{023} identify two as n, n' and force the third, n'', in precisely the positions of (5). Intersecting with $G_{13} = \mathrm{Sz}(u^d)$ proves the final divisibility condition. The inherited unique torus determines its compatible branch. Its containment in the ambient torus also gives the asserted ambient divisibility. □

5 The pure subfield residue R13

Theorem 7. *Suppose $G_0 = S_{N/d}$ is maximal in S_N and its prescribed rank-three residue has parabolics $G_{0i} = S_{N/(da_i)}$ for $i = 1, 2, 3$, where a_1, a_2, a_3 are distinct odd primes and their common intersection is simple. Every rank-four residually weakly primitive extension has $a_i \neq d$ and $G_i = S_{N/a_i}$, after one simultaneous conjugation. All intersections are the standard subfield intersections.*

Proof. Each G_i contains the simple G_{0i}, hence is a simple subfield group. The common simple subgroup standardizes every one of them by Lemma 1. Strictness excludes $G_i \leq G_0$. Lemma 3 now gives the stated degrees. Their simultaneous standard coordinates give every intersection by the gcd formula. □

6 Source scope and attribution

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, fixed edition, pp. 14–15. The R10 paragraph interchanges the labels of its second new Suzuki parabolic and its dihedral parabolic. The consistent labels are G_1, G_3 for the Suzuki groups and G_2 for the dihedral group, as fixed by the prescribed intersections in its opening sentence.

The ambient structure and torus compatibility are borrowed as stated. The overgroup, divisor-lattice and conditional extension arguments above are supplied here to repair a gap in our first claim ledger. A refutation of the source's global classification did not settle these distinct conditional assertions. The present theorems settle the rank-four assertions for the actual prescribed residues. The accepted constructions already prove realization of their admissible canonical parameters. Neither those existence results nor this note classifies all unrestricted higher-rank relative positions. In particular, the source's separate upper rank bound for pure subfield constructions remains refuted by the previously accepted repeated-prime example. No new open-conjecture, principal-refutation, Lean or compression credit is asserted here.